

# **B.Sc. CHEMISTRY SYLLABUS**

**(Common Core Syllabus Under CBCS)**

**With Effect From The Academic Year 2025-26 Onwards**



*Accredited with 'A<sup>+</sup>' by NAAC*

**Department of Chemistry**

**Kakatiya University**

**Warangal, Telangana- 506009**

Department of Chemistry, Kakatiya University, Warangal

# Chemistry Course Structure

KAKATIYA UNIVERSITY, WARANGAL.

B.Sc. CBCS COMMON CORE SYLLABUS FOR ALL UNIVERSITIES IN TELANGANA STATE

APPROVED SCHEME FOR CHOICE BASED CREDIT SYSTEM IN

**B.Sc. CHEMISTRY**

(Effect from the academic year 2025 – 2026 onwards)

| <b>FIRST YEAR-SEMESTER I</b>    |                                                                                                                          |                    |            |                |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------|------------|----------------|
| <b>CODE</b>                     | <b>COURSE TITLE</b>                                                                                                      | <b>COURSE TYPE</b> | <b>HPW</b> | <b>CREDITS</b> |
| BS 101                          | <b>Major – 1: Chemistry – I<br/>Laboratory Course – I<br/>( Quantitative Analysis – Titrations )</b>                     | DSC-1A             | 4T+2P = 6  | 4+1 = 5        |
| BS 102                          | Major – 2                                                                                                                | DSC-2A             | 4T+2P = 6  | 4+1 = 5        |
| BS 103                          | Minor                                                                                                                    | DSC-3A             | 4T+2P = 6  | 4+1 = 5        |
| BS 104                          | English                                                                                                                  | AEC-1A             | 5          | 5              |
| BS 105                          | Second Language                                                                                                          | AEC-2A             | 5          | 5              |
|                                 | <b>TOTAL CREDITS</b>                                                                                                     |                    | <b>28</b>  | <b>25</b>      |
| <b>FIRST YEAR-SEMESTER II</b>   |                                                                                                                          |                    |            |                |
| BS 201                          | <b>Major – 1: Chemistry – II<br/>Laboratory Course - II<br/>(Qualitative Analysis - Semi Micro Analysis of Mixtures)</b> | DSC-1B             | 4T+2P = 6  | 4+1 = 5        |
| BS 202                          | Major – 2                                                                                                                | DSC-2B             | 4T+2P = 6  | 4+1 = 5        |
| BS 203                          | Minor                                                                                                                    | DSC-3B             | 4T+2P = 6  | 4+1 = 5        |
| BS 204                          | English                                                                                                                  | AEC-1B             | 5          | 5              |
| BS 205                          | Second Language                                                                                                          | AEC-2B             | 5          | 5              |
|                                 | <b>TOTAL CREDITS</b>                                                                                                     |                    | <b>28</b>  | <b>25</b>      |
| <b>SECOND YEAR-SEMESTER III</b> |                                                                                                                          |                    |            |                |
| <b>CODE</b>                     | <b>COURSE TITLE</b>                                                                                                      | <b>COURSE TYPE</b> | <b>HPW</b> | <b>CREDITS</b> |
| BS 301                          | <b>Major – 1: Chemistry – III<br/>Laboratory Course - III<br/>(Synthesis of Organic compounds )</b>                      | DSC-1C             | 4T+2P = 6  | 4+1 = 5        |
| BS 302                          | Major – 2                                                                                                                | DSC-2C             | 4T+2P = 6  | 4+1 = 5        |
| BS 303                          | Minor                                                                                                                    | DSC-3C             | 4T+2P = 6  | 4+1 = 5        |
| BS 304                          | English                                                                                                                  | AEC-1C             | 5          | 5              |
| BS 305                          | Second Language                                                                                                          | AEC-2C             | 5          | 5              |
|                                 | <b>TOTAL CREDITS</b>                                                                                                     |                    | <b>28</b>  | <b>25</b>      |
| <b>SECOND YEAR-SEMESTER IV</b>  |                                                                                                                          |                    |            |                |
| BS 401                          | <b>Major – 1: Chemistry – IV<br/>Laboratory Course - IV<br/>(Qualitative Analysis of Organic Compounds)</b>              | DSC-1D             | 4T+2P = 6  | 4+1 = 5        |
| BS 402                          | Major – 2                                                                                                                | DSC-2D             | 4T+2P = 6  | 4+1 = 5        |
| BS 403                          | Minor                                                                                                                    | DSC-3D             | 4T+2P = 6  | 4+1 = 5        |
| BS 404                          | English                                                                                                                  | AEC-1D             | 5          | 5              |
| BS 405                          | Second Language                                                                                                          | AEC-2D             | 5          | 5              |
|                                 | <b>TOTAL CREDITS</b>                                                                                                     |                    | <b>28</b>  | <b>25</b>      |

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| <b>THIRD YEAR-SEMESTER V</b>  |                                                                                                                                                                                                                                          |                    |            |                |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------|----------------|
| <b>CODE</b>                   | <b>COURSE TITLE</b>                                                                                                                                                                                                                      | <b>COURSE TYPE</b> | <b>HPW</b> | <b>CREDITS</b> |
| BS 501                        | <b>Major – 1: Chemistry – V A/B</b><br><b>A. Spectroscopy and Chromatography</b><br><b>(Or)</b><br><b>B. Metallurgy, Dyes, Batteries and Nanomaterials</b><br><b>Laboratory Course – V</b><br><b>(Experiments in Physical Chemistry)</b> | DSC-1E             | 4T+2P = 6  | 4+1 = 5        |
| BS 502                        | Major – 2                                                                                                                                                                                                                                | DSC-2E             | 4T+2P = 6  | 4+1 = 5        |
| BS 503                        | <b>MDC: Foundations of Chemistry – Concepts and Applications</b>                                                                                                                                                                         | MDC-1              | 4          | 4              |
| BS 504                        | Skill Enhancement Course – 1:                                                                                                                                                                                                            | SEC-1              | 2          | 2              |
| BS 505                        | Skill Enhancement Course – 2:                                                                                                                                                                                                            | SEC-2              | 2          | 2              |
| BS 506                        | Value Added Course - 1                                                                                                                                                                                                                   | VAC-1              | 3          | 3              |
|                               | <b>TOTAL CREDITS</b>                                                                                                                                                                                                                     |                    | <b>23</b>  | <b>21</b>      |
| <b>THIRD YEAR-SEMESTER VI</b> |                                                                                                                                                                                                                                          |                    |            |                |
| BS 601                        | <b>Major – 1: Chemistry–VI A/B</b><br><b>A. Medicinal Chemistry</b><br><b>(Or)</b><br><b>B. Agricultural and Fuel Chemistry</b><br><b>Laboratory Course - VI</b><br><b>(Experiments in Physical Chemistry)</b>                           | DSC-1F             | 4T+2P = 6  | 4+1 = 5        |
| BS 602                        | Major – 2 (A/B)                                                                                                                                                                                                                          | DSC-2F             | 4T+2P = 6  | 4+1 = 5        |
| BS 603                        | Skill Enhancement Course – 3:                                                                                                                                                                                                            | SEC-3              | 2          | 2              |
| BS 604                        | Skill Enhancement Course – 4:                                                                                                                                                                                                            | SEC-4              | 2          | 2              |
| BS 605                        | Value Added Course - 2                                                                                                                                                                                                                   | VAC-2              | 3          | 3              |
| BS 606                        | Internship/ Project Work                                                                                                                                                                                                                 |                    | 4          | 4              |
|                               | <b>TOTAL CREDITS</b>                                                                                                                                                                                                                     |                    | <b>23</b>  | <b>21</b>      |
|                               | <b>GRAND TOTAL</b>                                                                                                                                                                                                                       |                    |            | <b>142</b>     |

**DSC:** Discipline Specific Course, **AEC:** Ability Enhancement Course, **MDC:** Multi-Disciplinary Course,

**SEC:** Skill Enhancement Course. **VAC:** Value Added Course

**Note:** Under the proposed CBCS structure for Undergraduate Programmes (2025-2026), students will pursue two Majors and one Minor, collectively referred to as Discipline Specific Course (DSC). Both Major and Minor courses will remain integrated up to the fourth semester, after which students will have the option to drop one Minor course in both the fifth and sixth semesters. The syllabus and credits for Major and Minor courses will remain consistent up to the fourth semester, with the Major continuing through the fifth and sixth semesters.

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**Credits under Non-CGPA  
(Community Engagement & Service)**

|    |                                |                                         |
|----|--------------------------------|-----------------------------------------|
| 1. | NSS/NCC/Sports/Extracurricular | Up to 6 Credits (2 in each year)        |
| 2. | IKS                            | Up to 4 (2 in each, after I & II years) |

**Credits under -CGPA**

|                                            |            |
|--------------------------------------------|------------|
| Major – 1                                  | 30         |
| Major – 2                                  | 30         |
| Minor                                      | 20         |
| AEC (Ability Enhancement Course) – English | 20         |
| Second Language                            | 20         |
| MDC (Multi-Disciplinary Course)            | 4          |
| SEC (Skill Enhancement Course)             | 8          |
| VAC (Value Added Course)                   | 6          |
| Project                                    | 4          |
| <b>Total</b>                               | <b>142</b> |

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**KAKATIYA UNIVERSITY**

**FACULTY OF SCIENCE**

**Theory Model Question Paper for End-Examinations-Semester I, II, III & IV**

**With Effect From the Academic Year 2025-2026 Onwards**

**B.Sc - CHEMISTRY (CBCS)**

**Time:3Hours**

**Max. Marks:80**

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**Part-A**

**(Short Answer Type)**

**I. Write any eight (8) questions of the following**

**(8x4=32Marks)**

- 1 }  
2 } Unit - I  
3 }  
4 }  
5 } Unit - II  
6 }  
7 }  
8 } Unit - III  
9 }  
10 }  
11 } Unit - IV  
12 }

**Part-B**

**(Essay Answer Type)**

**II. Answer all Questions**

**(4x 12= 48 Marks)**

- 13 a) }  
OR } Unit - I  
b) }
- 14 a) }  
OR } Unit - II  
b) }
- 15 a) }  
OR } Unit - III  
b) }
- 16 a) }  
OR } Unit - IV  
b) }

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**KAKATIYA UNIVERSITY**  
**FACULTY OF SCIENCE**  
**Scheme of Internal Assessment-I- from the Academic Year 2025-2026 Onwards**

**[DSC Subjects only]**

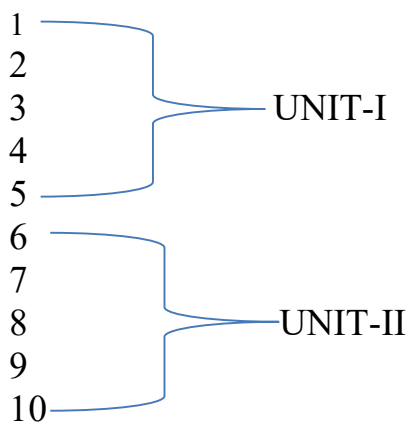
**Course : B. Sc : Chemistry**

**Time: 90 Min.**

**Max. Marks: 20**

**Answer ALL Questions. Each Question Carries Equal Marks**

**(10 × 2 = 20 Marks)**



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**KAKATIYA UNIVERSITY**

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**Scheme of Internal Assessment-II- from the Academic Year 2025-2026 Onwards**

**[DSC Subjects only]**

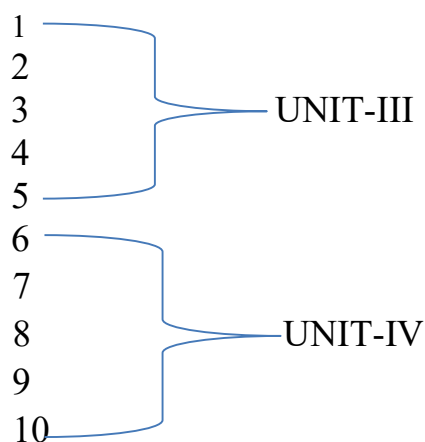
**Course : B. Sc- Chemistry**

**Time: 90 Min.**

**Max. Marks: 20**

**Answer ALL Questions. Each Question Carries Equal Marks**

**(10 × 2 = 20 Marks)**



**KAKATIYA UNIVERSITY**  
**FACULTY OF SCIENCE**  
**B. Sc-CHEMISTRY (CBCS)**

**Practical Examination Marks & Credits**

| <b>Semester</b> | <b>Time Duration</b> | <b>Total (Marks)</b> | <b>Credits</b> |
|-----------------|----------------------|----------------------|----------------|
| <b>I</b>        | <b>3 Hours</b>       | <b>25</b>            | <b>1</b>       |
| <b>II</b>       | <b>3 Hours</b>       | <b>25</b>            | <b>1</b>       |
| <b>III</b>      | <b>3 Hours</b>       | <b>25</b>            | <b>1</b>       |
| <b>IV</b>       | <b>3 Hours</b>       | <b>25</b>            | <b>1</b>       |

**B.Sc. I YEAR CHEMISTRY**  
**SEMESTER- I**  
**Paper – I**  
**Chemistry-I**

**Unit-I (Inorganic Chemistry)**

**15h (1h/week)**

**S1-I-1: Chemistry of P-Block Elements**

**15h**

Structure and bonding in diborane ( $B_2H_6$ ), Boron nitrogen compounds ( $B_3N_3H_6$  and BN), Lewis acid nature of  $BX_3$ .

**Carbides**- Classification -ionic, covalent, interstitial-Structures and reactivity. Industrial applications. Silicones-Classification-straight chain, cyclic and cross-linked and applications.

**Nitrides**-Classification -ionic, covalent and interstitial- Reactivity – hydrolysis.

**Oxides and Oxiacids :** Definition and Types of oxides (a) Normal-acidic, basic amphoteric and neutral (b) Mixed oxide (c) suboxide (d) peroxide (e) superoxide. Structure of oxides and oxy acids of B, C, N, P, S and Cl-reactivity, thermal stability, hydrolysis.

**Interhalogens**-Classification-general preparation-structures of  $AB$ ,  $AB_3$ ,  $AB_5$  and  $AB_7$  type and reactivity.

**Poly halide:** Definition and structure of  $ICl_2^-$ ,  $ICl_4^-$  and  $I_3^-$ .

**Pseudo halogens:** Comparison with halogens.

**Structure, bonding and reactivity of Xenon Compounds**- Oxides, Halides and Oxy-halides.

**Unit-II (Organic Chemistry)**

**15h (1h/week)**

**S1-O-1: Structural Theory in Organic Chemistry**

**5h**

**Bond polarization:** Factors influencing the polarization of covalent bonds, electro negativity – inductive effect. Application of inductive effect (a) Basicity of amines (b) Acidity of carboxylic acids (c) Stability of carbonium ions. Resonance - Mesomeric effect, application to (a) acidity of phenol. (b) acidity of carboxylic acids and basicity of anilines. Stability of carbo cations, carbanions and free radicals. Hyper conjugation and its application to stability of carbonium ions, free radicals and alkenes

**S1-O-2: Acyclic Hydrocarbons**

**5h**

**Alkanes** – Methods of preparation: Preparation of Alkanes from Grignard reagent. Chemical reactivity- inert nature, free radical substitution, Halogenation example.

**Alkenes**-Preparation of alkenes(with mechanism) (a) by dehydration of alcohols

(b) dehydrohalogenation of alkyl halides (c) by dehalogenation of 1,2 dihalides, Zaitsev's rule.

**Properties:** Anti-addition of halogen and its mechanism. Addition of  $HX$ , Markovnikov's rule, addition of  $H_2O$ ,  $HOX$  with mechanism and addition of  $HBr$  in the presence of peroxide (anti-Markovnikov's addition). Oxidation(cis-additions) hydroxylation by  $KMnO_4$ ,  $OsO_4$ , anti-addition-peracids (via epoxidation), ozonolysis – location of double bond.

**Alkynes**– Preparation by dehydrohalogenation of vicinal dihalides, dehalogenation of tetrahalides.

**Physical Properties:** Chemical reactivity – electrophilic addition of  $X_2$ ,  $HX$ ,  $H_2O$  (tautomerism), Oxidation (formation of enediol) and reduction (catalytic hydrogenation).

**S1-O-3: Aromatic Hydrocarbons**

**5h**

**Introduction to aromaticity:** Huckel's rule – Benzene, Naphthalene and Anthracene. Reactions - General mechanism of electrophilic substitution, mechanism of nitration, sulphonation and

halogenation, Friedel Crafts alkylation and acylation. Orientation of aromatic substitution - Definition of ortho, para and meta directing groups. Ring activating and deactivating groups with examples. Orientation – (i) activating groups: Amino, methoxy and alkyl groups. (ii) Deactivating groups - nitro, nitrile, carbonyl, carboxylic acid, sulphonic acid and halo groups.

### **Unit–III (Physical Chemistry)**

**15h (1h/week)**

#### **S1-P-1:Elementary quantum mechanics**

**3h**

Limitations of classical mechanics and Origin of quantum mechanics-Black body radiation, Rayleigh Jeans law; Planck's radiation law, photo electric effect, Compton effect, de Broglie's hypothesis. Heisenberg's uncertainty principle. Schrödinger wave equation (derivation not required) – significance of  $\psi$  and  $\psi^2$ .

#### **S1-P-2:Chemical Kinetics**

**8h**

Introduction to chemical kinetics, rate of reaction, rate laws and rate constant. Molecularity and Order of a reaction. Factors influencing the reaction rates. First order reaction, derivation of equation for rate constant. Characteristics of first order reaction. Units for rate constant. Half- life period, graph of first order reaction, Example - Decomposition of  $\text{H}_2\text{O}_2$  problems. Pseudo first order reaction,Hydrolysisofmethylacetate,inversionofcanesugar,problems.

Secondorderreaction,derivationofexpressionforsecondorderrateconstant,example-Saponification of ester. Characteristics of second order reaction, units for rate constants, half- life period and second order plots. Problems. Methods for determining the order of a reaction. Arrhenius equation – activation energy -problems.

#### **S1-P-3: Photochemistry**

**4h**

Introduction to photochemistry – differences between dark and photo reactions. Laws of photochemistry; Quantum Yield – problems; Examples of photo chemical reactions with different quantum yields. Photo chemical combinations of  $\text{H}_2\text{--Cl}_2$  and  $\text{H}_2\text{--Br}_2$  reactions.Abnormal quantum yield – high and low-examples with reasons. Singlet and triplet states. Jablonski diagram – non-radiative processes – Internal conversion and Intersystem crossing; radiative processes- Fluorescence and phosphorescence.

### **Unit-IV (General Chemistry)**

**15h (1h/week)**

#### **S1-G-1.General Principles of Inorganic quantitative Analysis:**

**5h**

**Volumetric Analysis:** Introduction, standard solutions, indicators, end point, titration curves, Types of titrations: i) neutralization titration- principle, theory of acid base indicators, titration curves and selection of indicators- strong acid - strong base, strong acid –weak base, weak acid-strong base and weak acid –weak base. Theory of redox titrations – internal ( $\text{KMnO}_4$ ) and external indicators – use of diphenylamine and ferroin indicators. Theory of complexometric titrations – use of EBT, Murexide and Fast sulphone black indicators. Role of pH in complexometric titrations. Precipitation titrations – theory of adsorption indicators.

#### **S1-G-2. Isomerism**

**5h**

**Isomerism:** Definition of isomers. Classification of isomers: Constitutional and Stereoisomers - definition and examples. Constitutional isomers: chain, functional and positional isomers.

Stereoisomers: enantiomers and diastereomers – definitions and examples. Representation of stereoisomers – Wedge, Fischer, Sawhorse, Newmann projection formulae.

**Conformational Analysis** :Classification of stereoisomers based on energy. Definition and examples Conformational and configurational isomers. Conformational analysis of ethane, n-butane, 1,2- dichloroethane, 2-chloroethanol. Cis-trans isomerism: E-Z-Nomenclature.

### **S1-G-3Colloids&Surface Chemistry**

**5h**

**Colloids:** Definition of colloids-classification of colloids-examples. Solid in liquid (sol)-Preparation, kinetic and electrical properties, stability and protection of colloids - Hardy-Schulze rule and Gold number. Liquid in liquid (emulsion)-types of emulsions and emulsifier. Liquid in solid (gel)-types and properties. Applications of colloids.

**Adsorption:** Types of adsorptions; Factors influencing adsorption; Freundlich adsorption isotherm and Langmuir adsorption isotherm. Applications.

### **References**

General reference: B. Sc I Year Chemistry: Semester I, Telugu Academy publication, Hyd.

#### **Unit-I**

1. Puri, B. R., Sharma, L. R., & Kalia, M. S. (1996). Principles of inorganic chemistry. Vishal Publications.
2. Lee, J. D. (1981). Concise inorganic chemistry (3<sup>rd</sup>ed.). Oxford University Press.
3. Cotton, F. A., Wilkinson, G., & Gaus, P. L. (2001). Basic inorganic chemistry (3<sup>rd</sup>ed.). Wiley.
4. Huheey, J. E., Keiter, E. A., & Keiter, R. L. (1993). Inorganic chemistry: Principles of structure and reactivity (4<sup>th</sup>ed.). Harper Collins College Publishers.
5. Greenwood, N. N., & Earnshaw, A. (1989). Chemistry of the elements. Pergamon Press.
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7. Gopalan, R. (2009). Textbook of inorganic chemistry. Universities Press.

#### **Unit-II**

1. Morrison, R. T., & Boyd, R. N. (2011). Organic chemistry. Pearson Education (Prentice Hall).
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3. Bruice, P. Y. (2017). Organic chemistry. Pearson Education.
4. Wade, L. G., Jr. (2013). Organic chemistry. Pearson Education.
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6. McMurry, J. (2015). Organic chemistry. Cengage Learning (Brooks/Cole).
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8. Ghosh, S. K. (2009). General organic chemistry. Bharati Bhawan Publishers.
9. Pillai, C. N. (2008). Organic chemistry. Universities Press (India) Pvt. Ltd.

#### **Unit III**

1. Puri, B. R., Sharma, L. R., & Pathania, M. S. (2013). Principles of physical chemistry (46<sup>th</sup>ed.). Vishal Publishing Company.
2. Raj, G. (2009). Advanced physical chemistry (35<sup>th</sup>ed.). Goel Publishing House.
3. Lewis, G., & Glasstone, S. (1966). Elements of physical chemistry. Macmillan.
4. Atkins, P. W. (2001). Physical chemistry (7<sup>th</sup>ed.). Oxford University Press.
5. Kapoor, K. L. (1994). A textbook of physical chemistry (Vols. 4 & 5). Macmillan India Ltd.
6. Laidler, K. J. (1987). Chemical kinetics (3<sup>rd</sup>ed.). McGraw Hill.
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8. Turro, N. J. (1978). Molecular photochemistry. W. A. Benjamin, Inc.  
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9. Rohatgi-Mukherjee, K. K. (1978). Fundamentals of photochemistry. Wiley Eastern.
10. Dogra, S. K., & Dogra, S. (1996). Physical chemistry through problems (4<sup>th</sup>ed.). New Age International.
11. Kalidas, C. & Sangaranarayanan, M. V. (2019). Physical chemistry: Problems and solutions. Universities Press.

#### Unit IV

1. Jeffery, G. H., Bassett, J., Mendham, J., & Denney, R. C. (1999). Vogel's textbook of quantitative chemical analysis (5<sup>th</sup>ed.). Addison Wesley Longman Inc.
2. Day, R. A., & Underwood, A. L. (2004). Quantitative analysis (6<sup>th</sup>ed.). Prentice Hall of India.
3. Svehla, G. (1996). Vogel's qualitative inorganic analysis (7<sup>th</sup>ed.). Prentice Hall.
4. Morrison, R. T., & Boyd, R. N. (2011). Organic chemistry. Pearson Education.
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9. Kapoor, K. L. (1994). A textbook of physical chemistry (Vols. 4 & 5). Macmillan India Ltd.
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11. McQuarrie, D. A., & Simon, J. D. (1997). Physical chemistry: A molecular approach. Viva Books Pvt. Ltd.
12. Satake, M., Hayashi, Y., Mido, Y., Iqbal, S. A., & Sethi, M. S. (2014). Colloidal and surface chemistry. Discovery Publishing Pvt. Ltd.

#### Laboratory Course-I

30h (2h/ week)

#### Paper-I: Quantitative Analysis

##### Acid-Base Titrations

1. Estimation of Carbonate in Washing Soda.
2. Estimation of Bicarbonate in Baking Soda.
3. Estimation of Carbonate and Bicarbonate in the Mixture.
4. Estimation of Alkali content in Antacid using HCl.

##### Redox Titrations

1. Determination of Fe(II) using  $K_2Cr_2O_7$
2. Determination of Fe(II) using  $KMnO_4$  with sodium oxalate as primary standard.
3. Determination of Cu(II) using  $Na_2S_2O_3$  with  $K_2Cr_2O_7$  as primary standard

##### Complexometric Titrations

1. Estimation of  $Mg^{2+}$  by EDTA
2. Estimation of  $Cu^{2+}$  by EDTA

#### References

1. Jeffery, G. H., Bassett, J., Mendham, J., & Denney, R. C. (1999). Vogel's textbook of quantitative chemical analysis (5<sup>th</sup>ed.). Addison Wesley Longman Inc.
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3. Giri, A. N. (2010). A textbook of practical chemistry. Himalaya Publishing House.
4. O.P. Pandey, D.N. Bajpai, & S. Giri. (2020). Practical chemistry. 10<sup>th</sup> Revised Edition, S. Chand Publishing. ISBN: 9789352535859
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- chemistry. Sultan Chand & Sons.
6. Gopalan, R, Venkappayya , D., and Nagarajan, S. (2012). Textbook of Inorganic chemistry (Lab Manual), 3<sup>rd</sup> Edition, Universities Press, Hyderabad, ISBN: 9788173718204
  7. Ahluwalia, V. K., and Sunita Dhingra, (2005). A Laboratory Manual of Organic and Inorganic Chemistry, 1<sup>st</sup> Edition, University Press, Hyderabad, ISBN: 9788173715623

**B. Sc I YEAR CHEMISTRY**  
**SEMESTER II**  
**Paper – II**  
**Chemistry–II**

**Unit-I (Inorganic Chemistry)**

**15h (1h/week)**

**S2-I-1 Chemistry of d-block elements**

**7h**

Characteristics of d-block elements with special reference to electronic configuration, variable oxidation states, color properties, d-d spectral transitions, ability to form complexes, magnetic properties, calculation of magnetic moment-spin only formula & catalytic properties. Comparative treatment of second and third transition series with their 3d analogues.

**S2-I-2: Chemistry of f-block elements**

**8h**

**Chemistry of Lanthanides:** Position in periodic table, electronic structure, oxidation state, ionic and atomic radii/ionic radii- lanthanide contraction- cause and consequences, anomalous behavior of post lanthanides-complexation-type of donor ligands preferred. Magnetic properties- paramagnetism, color and spectra, f-f transitions-occurrence and separation-ion exchange method, solvent extraction.

**Chemistry of actinides:** General features-electronic configuration, oxidation state, actinide contraction, color and complex formation. Comparison with lanthanides.

**Unit-II (Organic Chemistry)**

**15h (1h/week)**

**S2-O-1: Halogen compounds**

**4h**

**Classification:** alkyl (primary, secondary, tertiary), aryl, aralkyl, Chemical reactivity - reduction, formation of RMgX, Nucleophilic substitution reactions – classification into  $S_N^1$  and  $S_N^2$ . Mechanism and energy profile diagrams of  $S_N^1$  and  $S_N^2$  reactions. Stereochemistry of  $S_N^2$  (Walden Inversion) 2-bromobutane,  $S_N^1$  (Racemization) 1-bromo-1-phenylpropane.

**S2-O-2: Hydroxy compounds and ethers**

**5h**

**Alcohols:** Preparation: 1°, 2° and 3° alcohols using Grignard reagent, Reduction of Carbonyl compounds, carboxylic acids and esters. Physical properties: H-bonding, Boiling point and Solubility. Reactions with Sodium, HX/ZnCl<sub>2</sub> (Lucas reagent), oxidation with conc. HNO<sub>3</sub> and Oppenauer oxidation (Mechanism).

**Phenols:** Preparation: (i) from diazonium salts of anilines and (ii) from benzene sulphonic acids. Properties: Acidic nature, formation of phenoxide and reaction with R-X, electrophilic substitution; halogenations, Reimer Tiemann reaction (Mechanism), Gattermann-Koch reaction, Schotten-Baumann reaction.

**Ethers:** Nomenclature, preparation by Williamson synthesis. Chemical properties – inert nature, action of conc. H<sub>2</sub>SO<sub>4</sub>.

**S2-O-3 Carbonyl compounds**

**6h**

Preparation of aldehydes & ketones from acid chlorides, nitriles and carboxylic acids. Special methods of preparing aromatic aldehydes and ketones by Oxidation of arenes. Physical properties—absence of Hydrogen bonding. Reactivity of the carbonyl groups in aldehydes and ketones. Chemical reactivity: Addition of (a) NaHSO<sub>3</sub> (b) HCN (c) RMgX (d) 2,4-DNP (Schiff base). Addition of H<sub>2</sub>O to form hydrate, addition of alcohols -hemiacetal and acetal formation. Cannizzaro reaction. Oxidation reactions – KMnO<sub>4</sub> oxidation, reduction – catalytic hydrogenation, mechanism of Clemmensen reduction, Meerwein-Ponndorf-Verley reduction.

**Unit-III (Physical Chemistry)****15h (1h/week)****S2-P-1:Electrochemistry****15h**

Revision of conductance, specific conductance, equivalent conductance and factors influencing conductance of electrolytes. Ionic mobility, definition and significance of transport number. Kohlrausch's law – its applications: determination of degree of dissociation and acid dissociation constant ( $K_a$ ) of weak acids, solubility product determination and conductometric titrations. Ostwald's dilution law – issues and limitations. Debye-Hückel -Onsager's equation for strong electrolytes (elementary treatment only).

**Types of electrodes** with examples – Types of reversible electrodes -the gas electrode, metal-metal ion, metal-insoluble salt, redox electrodes and ion-selective electrode. Reversible and irreversible cells; Nernst equation – EMF of a cell; representation of a cell-problems; electrode potentials-electrochemical series and its significance. Determination of pH – using quinhydrone and glass electrodes. Potentiometric titrations.

**Unit-IV (General Chemistry)****15h (1h/week)****S2-G-1:Chemical Bonding****5h**

Molecular orbital theory: Shapes and sign convention of atomic orbitals. Modes of bonds. Criteria for orbital overlap. LCAO concept.  $\pi$  and  $\sigma$  overlapping. Concept of Types of molecular orbitals: bonding, antibonding and non-bonding. MOED of homonuclear diatomic molecules -  $H_2$ ,  $N_2$ ,  $O_2$ ,  $O_2^-$ ,  $O_2^{2-}$ ,  $F_2$  (unhybridized diagrams only) and heteronuclear diatomics - CO,  $CN^-$ , NO,  $NO^+$  and HF, their bond order, stability and magnetic properties.

**S2-G-2: Stereoisomerism****5h**

**Optical activity:** Definition, wave nature of light, plane polarized light, optical rotation and specific rotation, chiral centers. Chiral molecules: definition and criteria-absence of plane, center and  $S_n$  axis of symmetry – asymmetric and dissymmetric molecules. Examples of asymmetric molecules (Glyceraldehyde, Lactic acid, Alanine) and dissymmetric molecules (trans-1,2-dichlorocyclopropane). Molecules with constitutionally symmetrical chiral carbons (Tartaric acid) Molecules with constitutionally unsymmetrical chiral carbons (2,3-dibromopentane). D, L configuration – examples. R, S – configuration: Cahn-Ingold-Prelog (CIP) rules.

**S2-G-3: Colligative Properties****5h**

Definition of colligative properties- relative lowering of vapour pressure-Raoult's law; Osmotic pressure; elevation of boiling point and depression of freezing point; thermodynamic relation between molecular weight and colligative property (derivations not required) -Problems.

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### Unit IV

1. Day, R. A., & Underwood, A. L. (2004). Quantitative analysis (6<sup>th</sup> ed.). Prentice Hall of India.
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4. Puri, B. R., Sharma, L. R., & Pathania, M. S. (2013). Principles of physical chemistry (46<sup>th</sup>ed.). Vishal Publishing Company.
5. Raj, G. (2009). Advanced physical chemistry (35<sup>th</sup>ed.). Goel Publishing House.

#### Laboratory Course-II

**30h (2h/ week)**

#### Paper II - Qualitative Analysis- Semi micro analysis of mixtures

Department of Chemistry, Kakatiya University, Warangal

Analysis of two anions (one simple, one interfering) and two cations in the given mixture.

**Anions:**  $\text{CO}_3^{2-}$ ,  $\text{SO}_3^{2-}$ ,  $\text{S}^{2-}$ ,  $\text{Cl}^-$ ,  $\text{Br}^-$ ,  $\text{I}^-$ ,  $\text{CH}_3\text{COO}^-$ ,  $\text{NO}_3^-$ ,  $\text{PO}_4^{3-}$ ,  $\text{BO}_3^{3-}$ ,  $\text{SO}_4^{2-}$

**Cations:**  $\text{Hg}_2^{2+}$ ,  $\text{Ag}^+$ ,  $\text{Pb}^{2+}$ ,

$\text{Hg}^{2+}$ ,  $\text{Bi}^{3+}$ ,  $\text{Cd}^{2+}$ ,  $\text{Cu}^{2+}$ ,  $\text{As}^{3+}$  /  $\text{As}^{5+}$ ,  $\text{Sb}^{3+}$  /  $\text{Sb}^{5+}$ ,  $\text{Sn}^{2+}$  /  $\text{Sn}^{4+}$ ,

$\text{Al}^{3+}$ ,  $\text{Cr}^{3+}$ ,  $\text{Fe}^{3+}$

$\text{Mn}^{2+}$ ,  $\text{Co}^{2+}$ ,  $\text{Ni}^{2+}$ ,  $\text{Zn}^{2+}$

$\text{Ca}^{2+}$ ,  $\text{Ba}^{2+}$ ,  $\text{Sr}^{2+}$

$\text{Mg}^{2+}$ ,  $\text{NH}_4^+$

## References

1. Svehla, G. (1996). Vogel's qualitative inorganic analysis (7<sup>th</sup>ed.). Prentice Hall.
2. Gopalan, R., Subramanian, P. S., & Raghavan, K. (2004). Elements of analytical chemistry. Sultan Chand & Sons.
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**B .Sc II YEAR CHEMISTRY**  
**SEMESTER III**  
**Paper-III**  
**Chemistry-III**

**Unit-I (Inorganic Chemistry)**

**15h (1h/week)**

**S3-I-1: Coordination Compounds-I**

**10h**

Simple in organic molecules and coordination complexes. Nomenclature – IUPAC rules, 1. Coordination number, coordination geometries of metal ions, types of ligands. 2. Brief review of Werner's theory, Sidgwick's electronic interpretation and EAN rule and their limitations. (Valence bond theory (VBT) – postulates and application to (a) tetrahedral complexes  $[\text{Ni}(\text{NH}_3)_4]^{2+}$ ,  $[\text{NiCl}_4]^{2-}$  and  $[\text{Ni}(\text{CO})_4]$  (b) Square planar complexes  $[\text{Ni}(\text{CN})_4]^{2-}$ ,  $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ,  $[\text{PtCl}_4]^{2-}$  (c) Octahedral complexes  $[\text{Fe}(\text{CN})_6]^{4+}$ ,  $[\text{Fe}(\text{CN})_6]^{3-}$ ,  $[\text{FeF}_6]^{4+}$ ,  $[\text{Co}(\text{NH}_3)_6]^{3+}$ ,  $[\text{CoF}_6]^{3-}$ . Limitations of VBT. 3. Isomerism in coordination compounds, stereo isomerism-(a) Geometrical isomerism in (i) Square planar metal complexes of the type  $[\text{MA}_2\text{B}_2]$ ,  $[\text{MA}_2\text{BC}]$ ,  $[\text{M}(\text{AB})_2]$ ,  $[\text{MABCD}]$ . (ii) Octahedral metal complexes of the type  $[\text{MA}_4\text{B}_2]$ ,  $[\text{M}(\text{AA})_2\text{B}_2]$ ,  $[\text{MA}_3\text{B}_3]$  using suitable examples, (b) Optical isomerism in (i). tetrahedral complexes  $[\text{MABCD}]$ , (ii). Octahedral complexes  $[\text{M}(\text{AA})_2\text{B}_2]$ ,  $[\text{M}(\text{AA})_3]$  using suitable examples. Structural isomerism: ionization, linkage, coordination ligand isomerism using suitable examples.

**S3-I-2: Metal Carbonyls and related compounds**

**2h**

Metal Carbonyls: Classification, Structural features of  $\text{Ni}(\text{CO})_4$ ,  $\text{Fe}(\text{CO})_5$ ,  $\text{Fe}_2(\text{CO})_9$ ,  $\text{Fe}_3(\text{CO})_{12}$  and  $\text{Cr}(\text{CO})_6$ -18 valence electron rule.

**S3-I-3: Organometallic Chemistry**

**3h**

Definition, nomenclature and classification of organometallic compounds. Methods of preparation, properties and applications of alkyl and aryl compounds of Li & Mg.

**Unit-II (Organic Chemistry)**

**15h (1h/week)**

**S3-O-1: Carboxylic acids**

**6h**

**Preparation :** (a) Hydrolysis of Nitriles, amides and esters. (b) Carbonation of Grignard reagents. Special methods of preparation of Aromatic Acids -Oxidation of Arenes. Physical properties- hydrogen bonding, dimeric association, Chemical properties – Reactions involving H, OH and COOH groups -salt formation, anhydride formation, Acid halide formation, Esterification (mechanism) & Amide formation. Reduction of acid to the corresponding primary alcohol - via ester or acid chloride. Arndt – Eistert synthesis, Halogenation by Hell – Volhard -Zelinsky reaction.

**S3-O-2: Nitro hydrocarbon**

**4h**

Preparation of Nitroalkanes. Reactivity-halogenation, reaction with  $\text{HNO}_2$  (Nitrous acid), Nef reaction, reduction. Aromatic Nitro hydrocarbons: Preparation of Nitrobenzene by Nitration. Physical properties, chemical reactivity–Reduction of Nitrobenzene in different media.

**S3-O-3: Amines****5h**

Amines: classification into 1°, 2°, 3° Amines and quaternary ammonium compounds. Preparative methods – Ammonolysis of alkyl halides, Gabriel synthesis, Hoffmann bromamide reaction (mechanism). Reduction of Amides and Schmidt reaction. Physical properties. Chemical Properties: a) Alkylation b) Acylation c) Carbylamine reaction d) Hinsberg separation. Reaction with Nitrous acid of 1°, 2°, 3° amines (aliphatic and aromatic). Electrophilic substitutions of Aromatic amines – Bromination and Nitration.

**Unit III (Physical Chemistry)****15h (1h/week)****S3-P-1: Solid state Chemistry****6h**

Classification of Crystalline Solids; Definition of space lattice, unit cell. Bravais Lattices and Seven Crystal systems (a brief review). Laws of Crystallography-(i) Law of Constancy of interfacial angles (ii) Law of Symmetry- Symmetry elements in crystals (iii) Law of rationality of indices.; X-ray diffraction by crystals; Derivation of Bragg's equation. Determination of structure of NaCl (Bragg's method and Powder method)

**S3-P-2: Catalysis****9h**

Definition of a catalyst and catalysis. Comparison of homogeneous and heterogeneous catalysis with specific examples. General characteristics of catalytic reactions.

Acid-base catalysis- Specific acid and general acid catalysis- Examples, Specific base and general base catalysis – Examples. Effect of pH on reaction rate of acid and base catalysed reactions.

Enzyme catalysis- Characteristics of enzyme catalysis, Examples: Factors affecting enzyme catalysis. Effect of temperature, pH, concentration and effect of inhibitor on enzyme catalysed reactions, Catalytic efficiency. Michaelis-Menten Equation (Derivation not required). Significance of Michaelis constant ( $K_m$ ) and maximum velocity ( $V_{max}$ ), Lineweaver-Burk plot.

**Unit-IV (General Chemistry)****15h (1h/week)****S3-G-1 Bioinorganic Chemistry:****5h**

Essential elements, biological significance of Na, K, Mg, Ca, Fe, Co, Ni, Cu, Zn and chloride ( $Cl^-$ ). Toxicity of Cd, As, Hg and Pb. Oxygen transport and storage: structure of hemoglobin, binding and transport of oxygen. Fixation of  $CO_2$  in photosynthesis.

**S3-G-2: Heterocyclic Compound****5h**

Introduction and definition: 5-membered ring compounds with one hetero atom Ex. pyrrole, furan and thiophene. Importance of ring systems – and numbering in heterocyclic compounds. Aromatic character, Resonance structures: Explanation of feebly acidic character of pyrrole, preparation of pyrrole, furan and thiophene Paal-Knorr synthesis. electrophilic substitution, halogenation, nitration and sulfonation. Basicity, aromaticity of pyridine – comparison with pyrrole –preparation by Hantzsch method and properties – reactivity towards nucleophilic substitution reaction – Chichibabin reaction.

**S3-G-3:Phase Rule****5h**

Statement and meaning of the terms – Phase, Component and Degrees of freedom, Gibb's Phase rule, phase equilibria of one component system – water system. Phase equilibria of two-component system – Solid-Liquid equilibria, simple eutectic –Pb-Ag system, desilverisation of lead.

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**General reference:** B.ScII Year Chemistry: Semester III, Telugu Academy publication, Hyd

**Unit-I**

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**Unit-II**

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#### Laboratory Course-III

##### Paper III (Organic Synthesis)

30h (2h/ week)

##### 1. Synthesis of Organic compounds:

Acetylation: Acetylation of salicylic acid.

Aromatic electrophilic substitution: Nitration: Preparation of nitro benzene.

Halogenation: Preparation of p-bromo acetanilide.

Oxidation: Preparation of benzoic acid from benzyl chloride.

Methylation: Preparation of - naphthyl methyl ether.

Condensation: Preparation of Benzilidene aniline and Benzaldehyde and aniline.

Diazotisation: Azocoupling of  $\beta$ -Naphthol.

##### 2. Microwave assisted synthesis of Aspirin –DEMO (demonstration only)

###### References

1. Vogel, A. I., Furniss, B. S., Hannaford, A. J., Smith, P. W. G., & Tatchell, A. R. (1989). Vogel's textbook of practical organic chemistry (5<sup>th</sup>ed.). Longman.
2. Ahluwalia, V. K., & Aggarwal, R. (2000). Comprehensive practical organic chemistry: Preparation and quantitative analysis. Universities Press.
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**B.Sc. II YEAR CHEMISTRY**  
**SEMESTER WISE SYLLABUS**  
**SEMESTER IV**  
**Paper-IV Chemistry-IV**

**Unit-I (Inorganic Chemistry)**

**15h (1h/week)**

**S4-I-1: Coordination Compounds–II**

**12h**

Crystal field theory (CFT)- Postulates of CFT, splitting patterns of d-orbitals in octahedral, tetrahedral, square planar with suitable examples. Crystal field stabilization energies and its calculations for various  $d^n$  configurations in octahedral complexes. High Spin and Low Spin complexes. Colour and Magnetic properties of transition metal complexes. Calculations of magnetic moments spin only formula. Detection of complex formation - basic principles of various methods- change in chemical properties, solubility, colour, pH, conductivity, magnetic susceptibility.

**S4-I-2: Hard and soft acids bases (HSAB):**

Classification, Pearson's concept of hardness and softness, application of HSAB principles – Stability of compounds / complexes, predicting the feasibility of reaction.

**S4-I-3: Stability of metal Complexes:**

**3h**

Thermodynamic and kinetic stability of transition of metal complexes. Stability of metal complexes –stepwise and overall stability constant and their relationship. Determination of composition of complex by Job's method and mole ratio method.

**Unit-II (Organic Chemistry)**

**15h (1h/week)**

**S4-O-1: Carbohydrates**

**8h**

Introduction: Classification and nomenclature. Monosaccharides: All discussion to be confined to (+) glucose as an example of aldo hexoses and (-) fructose as example of ketohexoses. Chemical properties: Evidences for straight chain penta-hydroxy aldehyde structure. Number of optically active isomers possible for the structure and configuration of glucose based on D-glyceraldehyde as primary standard (No proof for configuration is required). Evidence for cyclic structure of glucose (Pyranose structure, anomeric Carbon and anomers). Proof for the ring size (methylation, hydrolysis and oxidation reactions). (Haworth formula and chair conformational formula). Structure of fructose: Evidence of 2 – ketohexose structure. Same osazone formation from glucose and fructose, Cyclic structure for fructose (Furanose structure, Haworth formula).

Inter Conversion of Monosaccharides: Arabinose to D-glucose, D- mannose (Kiliani – Fischer method). D-glucose to D-arabinose by Ruff degradation. Aldohexose(+) (glucose) to ketohexose (–) (fructose) and Ketohexose(Fructose) to aldohexose (Glucose).

**S4-O-2: Amino acids and proteins**

**7h**

Classification. Methods of synthesis: General methods of synthesis of alpha amino acids (specific examples – Glycine, Alanine, and Valine) by following methods: a) From halogenated Carboxylic acid b) Malonic ester synthesis. Physical properties: Optical  
Department of Chemistry, Kakatiya University, Warangal

activity of naturally occurring amino acids. Zwitter ion structure – salt like character, definition of isoelectric point. Chemical properties: General reactions due to amino and carboxyl groups – Lactam from gamma and delta amino acids by heating peptide bond (amide linkage). Structure and nomenclature of peptides.

### **Unit III (Physical Chemistry)**

**15h (1h/week)**

#### **S4-P-1: Thermodynamics**

**15h**

Revision of terms of thermodynamics: I law of thermodynamics-statements-concepts of Internal energy, enthalpy, heat capacity, work and heat. Expression for work done in isothermal and adiabatic processes -reversible ( $PV^\gamma = \text{constant}$ ) and irreversible processes – problems. Heat capacities at constant pressure and volume. Derivation of  $C_p - C_v = R$ . Limitations of I law-scope of II law- statements of second law of thermodynamics. Spontaneous and non-spontaneous processes, spontaneity and equilibrium. Cyclic process - Carnot cycle- derivation of efficiency based on entropy concept – problems. Physical significance of entropy. Change in entropy of an ideal gas as a function of P, V and T. Entropy changes of an ideal gas in various processes. Entropy as a criterion for spontaneity. Free energy–Gibb's free energy & Helmholtz free energy (work function), relation between w and  $\Delta A$  and  $\Delta G$ . Free energy -Variation of G with T,V and P- problems. Derivation of equation  $\Delta G = \Delta H - T\Delta S$ .  $\Delta G$  as criteria of equilibria or spontaneity of a reaction.

### **Unit IV (General Chemistry)**

**15h (1h/week)**

#### **S4-G-1: Evaluation of analytical Data**

**4h**

Significant figures , accuracy and precision. Errors-classification of errors-determinate and indeterminate errors, absolute and relative errors. Problems based on mean, median, range, standard deviation

#### **S4-G-2: Carbanions**

**5h**

Introduction, acidic nature of  $\alpha$ -hydrogens and tautomerism in carbonyl compounds, nitro hydrocarbons, ethyl acetoacetate, diethyl malonate. Terminal alkynes. Stability of carbanions Reactions: Aldol reaction, Perkin reaction, Benzoin condensation, haloform reaction, conversion of smaller alkynes to higher alkynes. Mannich reaction, Michael addition and Knoevenagel condensation Synthetic applications of Aceto acetic ester. Acid hydrolysis and ketonic hydrolysis.

#### **S4-G-3: AI (Artificial Intelligence) applications in Chemistry**

**6h**

##### **Introduction to AI :**

Definition of AI; Machine Learning (ML) and Types: Supervised, Unsupervised, Reinforcement Learning, AI vs. Traditional Programming. (Only conceptual information, very brief). Examples in daily life (voice assistants, image recognition)

##### **AI Tools in Chemistry :**

Usage of AI in Chemistry - Prediction of physical properties (solubility, pKa, boiling points) ; Molecular property prediction (toxicity, bioactivity)

Chemical Data Formats: Datasets and Descriptors Chemical data- SMILES, InChI, Molecular descriptors (size, shape, polarity),

Popular Free Chemical Databases: PubChem, ChemSpider, ChEMBL AI in drug design

### **Basics of Using AI Tools**

Introduction to user-friendly tools like: ACD/Labs, Molecule Net, ChemRxiv AI, ChatGPT for quick analysis Case study: Predicting solubility or boiling point of compounds (e.g., ethanol. Caffeine) using a free ML tool

### **Ethics, Limitations & Future Scope in Chemistry**

Limitations of AI in science, Ethics in data and model predictions

### **References:**

**General reference:** B.Sc II Year Chemistry: Semester IV, Telugu Academy publication, Hyd.

### **Unit-I**

1. Puri, B. R., Sharma, L. R., & Kalia, M. S. (1996). Principles of inorganic chemistry. Vishal Publications.
2. Lee, J. D. (1977). Concise inorganic chemistry (3<sup>rd</sup>ed.). Van Nostrand Reinhold Company.
3. Cotton, F. A., Wilkinson, G., & Gaus, P. L. (2001). Basic inorganic chemistry (3<sup>rd</sup>ed.). Wiley Publishers.
4. Huheey, J. E., Keiter, E. A., & Keiter, R. L. (2006). Inorganic chemistry: Principles of structure and reactivity (4<sup>th</sup>ed.). Pearson Education.
5. Greenwood, N. N., & Earnshaw, A. (1989). Chemistry of the elements. Pergamon Press.
6. Shriver, D. F., & Atkins, P. W. (1999). Inorganic chemistry (3<sup>rd</sup>ed.). Oxford University Press.

### **Unit-II**

1. Soni, P. L. (2012). Textbook of organic chemistry (29<sup>th</sup>ed.). Sultan Chand & Sons.
2. Ghosh, S. K. (2008). General organic chemistry. New Age Publishers Pvt. Ltd.
3. Morrison, R. T., & Boyd, R. N. (2009). Organic chemistry (6<sup>th</sup>ed.). Pearson Education.
4. Solomons, T. W. G., & Fryhle, C. B. (2015). Organic chemistry (11<sup>th</sup>ed.). Wiley.
5. Bruice, P. Y. (2012). Organic chemistry (7<sup>th</sup>ed.). Pearson Education.
6. Pillai, C. N. (2012). Textbook of organic chemistry. CRC Press.
7. Wade, L. G. Jr. (2013). Organic chemistry (8<sup>th</sup>ed.). Pearson Education.
8. Jones, M. Jr. (2010). Organic chemistry (4<sup>th</sup>ed.). W. W. Norton & Company.
9. McMurry, J. (2015). Organic chemistry (9<sup>th</sup>ed.). Cengage Learning.

### **Unit III**

1. Maron, S. H., & Lando, J. B. (1966). Fundamentals of physical chemistry. Macmillan Limited.
2. Puri, B. R., Sharma, L. R., & Pathania, M. S. (2013). Principles of physical chemistry (46<sup>th</sup> ed.). Vishal Publishing Company.
3. Raj, G. (2009). Advanced physical chemistry (35<sup>th</sup>ed.). Goel Publishing House.
4. Atkins, P. W. (2001). Physical chemistry (7<sup>th</sup>ed.). Oxford University Press.
5. Dogra, S. K., & Dogra, S. (1996). Physical chemistry through problems (4<sup>th</sup>ed.). New Age Int..
6. Castellan, G. W. (1985). Physical chemistry (3<sup>rd</sup>ed.). Narosa Publishing House.
7. Klotz, I. M., & Rosenberg, R. M. (1994). Chemical thermodynamics. John Wiley & Sons.
8. Rajaram, J., & Kuriacose, J. C. (1986). Thermodynamics. Shoban Lal Nagin Chand & Co.
9. Kapoor, K. L. (1994). A textbook of physical chemistry (Vol. 2). Macmillan India Ltd.
10. Sharma, K. K., & Sharma, L. K. (2012). A textbook of phy. Chem. (5<sup>th</sup>ed.). Vikas Pub.House.

## Unit IV

1. Puri, B. R., Sharma, L. R., & Kalia, M. S. (1996). Principles of inorganic chem. Vishal Pub.
2. Lee, J. D. (1977). Concise inorganic chemistry (3<sup>rd</sup>ed.). Van Nostrand Reinhold Company.
3. Morrison, R. T., & Boyd, R. N. (2009). Organic chemistry. Pearson Education.
4. Solomons, T. W. G., & Fryhle, C. B. (2015). Organic chemistry. Wiley.
5. Singh, J., & Yadav, L. D. S. (2010). Organic synthesis. Pragati Prakashan.
6. Clayden, J., Greeves, N., Warren, S., & Wothers, P. (2001). Orga. chemistry. Oxford Univ. Press.
7. Web tools: IBM RXN for Chemistry – AI-based reaction prediction tool Link: <https://rxn.app.accelerate.science>
8. ACD/Labs ChemSketch (Freeware) – Draw molecules, get properties, IUPAC names Link: <https://www.acdlabs.com/resources/free-chemistry-software-apps/chemsketch-freeware>
9. MoleculeNet (AI-ready datasets for chemical properties) Website: <https://moleculenet.org>
10. ChemRxiv (Chemistry Preprint Server) Website: <https://chemrxiv.org>
11. ChatGPT (AI chatbot for Q&A and explanations) Website: <https://chat.openai.com>
12. PubChem Website: <https://pubchem.ncbi.nlm.nih.gov>
13. ChemSpider Website: <http://www.chemspider.com> Maintained by: The Royal Society of Chemistry
14. ChEMBL Website: <https://www.ebi.ac.uk/chembl> Maintained by: The European Bioinformatics Institute (EMBL-EBI)

## Laboratory Course –IV

### Qualitative Analysis of Organic Compounds:

30h (2h/week)

Qualitative analysis: Identification of unknown organic compounds through the functional group analysis - ignition test, solubility test, functional group tests and preparation of suitable derivatives of the following: Carboxylic acids, phenols, amines, carbohydrates, aldehydes, ketones, ester and naphthalene.

## References

1. Furniss, B. S., Hannaford, A. J., Smith, P. W. G., & Tatchell, A. R. (1989). Vogel's textbook of practical organic chemistry (5<sup>th</sup>ed.). Longman.
2. Ahluwalia, V. K., & Aggarwal, R. (2000). Comprehensive practical organic chemistry: Qualitative analysis. Universities Press.
3. Mann, F. G., & Saunders, B. C. (1978). Practical organic chemistry (4<sup>th</sup>ed.). Cambridge University Press.
4. Bahl, A., & Bahl, B. S. (2009). Advanced practical organic chemistry. S. Chand Publishing.
5. Giri, A. N. (2010). A textbook of practical chemistry. Himalaya Publishing House.
6. Venkateswarlu P. (2023) Comprehensive Approach to Practical Organic Chemistry (Qualitative Analysis, Synthesis and UV, IR, NMR & MS Spectral Identification) by Pharma Med PRESS, BSPB Pvt. L, An Imprint of BSP Books Pvt. Ltd.

**Annexure – I (Credits)**  
**Proposed CBCS Structure from 2025-26 for Under Graduate Courses**

| Courses                                                         |                         | Papers                                       | Total Credits | Credits for each paper / Semester      |    |     |    |    |    | Credits for each paper / Semester      |    |     |    |    |    | Credits for each paper / Semester      |    |     |    |    |    |
|-----------------------------------------------------------------|-------------------------|----------------------------------------------|---------------|----------------------------------------|----|-----|----|----|----|----------------------------------------|----|-----|----|----|----|----------------------------------------|----|-----|----|----|----|
|                                                                 |                         |                                              |               | BA                                     |    |     |    |    |    | B.Com.                                 |    |     |    |    |    | B.Sc.                                  |    |     |    |    |    |
|                                                                 |                         |                                              |               | I                                      | II | III | IV | V  | VI | I                                      | II | III | IV | V  | VI | I                                      | II | III | IV | V  | VI |
| Core Courses<br>DSC                                             | Major-1                 | 6                                            | 30            | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  |
|                                                                 | Major -2                | 6                                            | 30            | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  |
|                                                                 | Minor-1                 | 4                                            | 20            | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  |
| MIL/AEC<br>(First Language)                                     | English                 | 4                                            | 20            | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  |
| Second Language<br>(Telugu, Hindi, Urdu, etc.)                  |                         | 4                                            | 20            | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  |
| Multi-<br>Disciplinary<br>Course                                | MDC 1                   | 1                                            | 4             | -                                      | -  | -   | -  | 4  | -  | -                                      | -  | -   | -  | 4  | -  | -                                      | -  | -   | -  | 4  | -  |
| Sec 1, 2                                                        |                         | 2                                            | 4             |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |
| Sec 3, 4                                                        |                         | 2                                            | 4             |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |
| Value added<br>course (VAC)                                     | VAC 1, 2                | 2                                            | 6             | -                                      | -  | -   | -  | 3  | 3  | -                                      | -  | -   | -  | 3  | 3  | -                                      | -  | -   | -  | 3  | 3  |
| Internships                                                     | Internship<br>/ Project | 1                                            | 4             | -                                      | -  | -   | -  | -  | 4  | -                                      | -  | -   | -  | -  | 4  | -                                      | -  | -   | -  | -  | 4  |
| Total Credits in each semester                                  |                         |                                              | 142           | 25                                     | 25 | 25  | 25 | 21 | 21 | 25                                     | 25 | 25  | 25 | 21 | 21 | 25                                     | 25 | 25  | 25 | 21 | 21 |
| Total Credits in UG                                             |                         |                                              |               | 142                                    |    |     |    |    |    | 142                                    |    |     |    |    |    | 142                                    |    |     |    |    |    |
| Credits under Non-CGPA<br>(Community engagement and<br>service) |                         | NSS /NCC<br>/sports /<br>Extra<br>curricular | 6             | Upto 6 (2 in each year)                |    |     |    |    |    | Upto 6 (2 in each year)                |    |     |    |    |    | Upto 6 (2 in each year)                |    |     |    |    |    |
|                                                                 |                         | IKS                                          | 4             | Upto 4 (2 in each, after I & II years) |    |     |    |    |    | Upto 4 (2 in each, after I & II years) |    |     |    |    |    | Upto 4 (2 in each, after I & II years) |    |     |    |    |    |



## **B.Sc., III YEAR CHEMISTRY**

### **SEMESTER-VI**

#### **DSE-A: Chemistry Paper-VI**

*(Medicinal Chemistry)*

**(04 credits)**

**60 Hrs (04 Hrs/week)**

#### **Unit- I: Introduction and Terminology (15 Hrs)**

**S6-E-A-I: Diseases:** Common diseases, infective diseases—insect borne, air-borne, water-borne and hereditary diseases.

**Terminology in Medicinal Chemistry:** Drug, Active Pharmaceutical Ingredient (ADI), Pharmaceuticals, Pharmacology, Pharmacophore, Pharmacodynamics, Pharmacokinetics, metabolites, anti metabolites and therapeutic index.

**Drugs:** Nomenclature: Chemical name, Generic name and Trade names with examples; Classification: Classification based on structures and therapeutic activity with examples.

**ADMET:** a) Absorption: Definition, absorption of drugs across the membrane – active and passive absorption, routes of administration of drugs. b) Distribution: definition and effect of plasma protein binding. c) Metabolism: definition, phase I and phase II reactions. d) Elimination: definition and renal elimination. Toxicity.

#### **Unit-II: Enzymes and Receptors (15 Hrs)**

**S6-E-A-II: Enzymes: Introduction,** Mechanism and factors affecting enzyme action, Specificity of enzyme action (including stereo specificity), Enzyme inhibitors and their importance. Types of inhibition - reversible, irreversible and their subtypes with examples.

**Receptors:** Introduction, Drug action-receptor theory, Mechanism of drug action, concept of agonists and antagonists with examples. Drug receptor interactions involved in drug receptor complex. Binding role of -OH group, -NH<sub>2</sub> group, quaternary ammonium salts and double bond. Structure – activity relationships of drug molecules, explanation with sulfonamides.

#### **Unit- III: Synthesis and Therapeutic Activity of Drugs (15 Hrs)**

**S6-E-A-III:** Introduction, synthesis and therapeutic activity of:

**Chemotherapeutics:** Sulphanilamide, dapsone, Penicillin-G (semi synthesis), Chloroquin, Isoniazid, Cisplatin and AZT.

**Drugs to treat metabolic disorders:** Anti diabetic - Tolbutamide; Anti-inflammatory – Ibuprofen; Cardiovascular- Glyceryl trinitrate; Antipyretic (paracetamol, aspirin) and Antacid- Omeprazole.

*For T.M.*

*Normeddy*

**Drugs acting on nervous system:** Anesthetics-definition, Classification-local and general. Volatile-Nitrous oxide, chloroform uses and disadvantages. Local anesthetics – benzocaine.

**Unit- IV: Molecular Messengers and Vitamins and Micronutrients (15 Hrs)**

**S6-E-A-IV: Molecular Messengers:** Introduction to hormones and neurotransmitters, Thyroid hormones, Antithyroid drug-Carbimazol. Adrenaline: Adrenergic drugs- salbutamol, atenelol. Serotonin: SSRIs- fluoxetine. Dopamine: Antiparkinson drug- Levodopa .

**Vitamins and Micronutrients:** Introduction, Vitamin sources, Deficiency disorders and remedy of Vitamins A,B, C, D, E, K and micronutrients – Na, K, Ca, Cu, Zn and I .

**Recommended Text Books and Reference Books:**

1. Introduction to Medicinal Chemistry, G.L. Patrick, Oxford University Press, New York. 2013.
2. Medicinal Chemistry, Thomas Nogrady, Oxford Univ. Press, New York.2005.
3. Foye's Principles of Medicinal Chemistry, David William and Thomas Lemke, Lippincott Williams & Wilkins, 2008.
4. Medicinal Chemistry, Ashutosh Kar, New Age International, 2005.
5. Synthetic Drugs, O.D.Tyagi & M.Yadav, Anmol Publications,1998.
6. Medicinal Chemistry, Alka L. Gupta, Pragati Prakashan.
7. Drugs, G. L. David Krupadanam, D.Vijaya Prasad, K.Varaprasad Rao, K. L. N. Reddy, C. Sudhakar, Universities Press (India) Ltd. 2012.







## B.Sc., III YEAR CHEMISTRY

### SEMESTER-VI

#### DSE-B: Chemistry Paper-VI

*(Agricultural & Fuel Chemistry)*

(04 credits)

60 Hrs (04 Hrs/week)

#### Unit I: Pesticides (15 Hrs)

**S6-E-B-I: Introduction**, Definition, classification of pesticides based on use (target). Toxicity and chemical structure with examples. Adverse effects of pesticides and its impact on environmental pollution.

Synthesis, manufacture and uses of representative pesticides: Organochlorines (Cypermethrin); Organophosphates (Parathion); Carbamates (carbaryl); Quinones (Chloranil), Anilides (Alachlor).

**Pesticide formulations**: Dusts, Granules, Wettable powders, Emulsions and Aerosols.

**Biopesticides** : Introduction: Potential pesticidal plants of India, Role of Neem in plant protection-constituents, Azadirachtin and its role in pest control, Structure and mode of action of Pyrethrins (pyrethrin-1) and Pyrethroids (permethrin) and nicotinoids (Imidacloprid).

#### Unit II: Fertilizers (15Hrs)

**S6-E-B-II: Introduction**: (need of fertilizers), functions of essential plant nutrients (N, P, K), Classification formula and uses of fertilizers:

**Nitrogenous fertilizers**: Ammonium nitrate, Urea, Calcium Cyanamide, Calcium Ammonium Nitrate, Sodium Nitrate, Ammonium Chloride and their uses.

**Phosphate fertilizers**: Normal super phosphate, Triple Super Phosphate, Ammonium Phosphate and their uses.

**Potassium fertilizers**: Potassium chloride, potassium nitrate, potassium sulphate and uses.

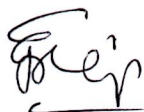
**Complex fertilizers**: Diaammonium Phosphate and mixed fertilizers their uses. Manufacture of urea and Super phosphate of lime and their reactions in the soil.

**Biofertilizers**: Introduction, definition, classification, Rhizobium, Azatobactor, Azospirillum, Azolla, Blue Green Algae, Vermicomposting and uses.

**Organic farming**: The principal methods, crop rotation, green manures and compost, biological pest control, and mechanical cultivation and uses.

#### Unit III: Energy Sources and Coal (15Hrs)

**S6-E-B-III**: Review of energy sources (renewable and non-renewable). Classification of fuels and their calorific value.



**Coal:** Uses of coal (fuel and nonfuel) in various industries, its composition, carbonization of coal. Coal gas, producer gas and water gas—composition and uses. Fractionation of coal tar, uses of coal tar based chemicals, requisites of a good metallurgical coke, Coal gasification (Hydro gasification and Catalytic gasification), Coal liquefaction and Solvent Refining.

#### **Unit IV: Petroleum and its products, Petrochemicals and non petroleum fuels (15Hrs)**

##### **S6-E-B-IV: Petroleum and its products**

**Petroleum:** Origin, Composition of crude petroleum and classification. Properties-flash point and its determination, Knocking and anti-knocking compounds; Octane number and Cetane number. Distillation of crude petroleum, Fractional Distillation - Principle and process, refining, fractions and uses. Cracking -Thermal and catalytic cracking, Reforming.

**Petroleum products** – Petrol, Diesel, LPG, Kerosene, Tar and their applications.

**Petrochemicals**-Vinyl acetate, Propylene oxide, Isoprene and their uses.

**Lubricants:** Classification of lubricants- Solid, semi solid and liquids; Properties (viscosity, flash point, fire point, cloud point, pour point) and their determination. Functions of Lubricants, Mechanism of lubrication.

**Non-Petroleum fuels:** Natural Gas- CNG, LNG, clean Fuels- H<sub>2</sub> gas, ethanol, Fuel from waste- bio gas, Fuel from bio mass-Bio ethanol, biodiesel, and Synthetic fuels- syngas based.

#### **Recommended Text Books and Reference Books:**

1. Chemistry of pesticides, N. N. Melnikov, Springer-Verlag- Technology & Engineering (2012).
2. Pesticide Synthesis, Thomas A. Unger, Elsevier, (2000).
3. Pesticides, R. Cremlyn, John Wiley, 1980.
4. Manures and Fertilisers, K. Kolay, Published by Atlantic (2007).
5. Sharma, B.K. & Gaur, H. Industrial Chemistry, Goel Publishing House, Meerut (1996).
6. A Text Book of Engineering Chemistry Paperback-2017 by Shashi Chawla.
7. Industrial Chemistry, Vol-I, Stocchi.E, Ellis Horwood Ltd. UK (1990).
8. Jain, P.C. & Jain, M. Engineering Chemistry, Dhanpat Rai & Sons, Delhi.
9. Engineering Chemistry by Shashi Chawla, Dhanpat Rai & Sons, Delhi.

*JP*

*Deep*

*ruddy*

**B.Sc., III YEAR CHEMISTRY**

**SEMESTER-VI**

**LABORATORY COURSE**

**Paper –V: Experiments in Physical Chemistry-II**

**(01 Credit)**

**45 Hrs (03 Hrs/week)**

**1. Kinetics**

- Determination of specific reaction rate of the hydrolysis of methyl acetate catalyzed by hydrogen ion at room temperature.
- Determination rate of decomposition of hydrogen peroxide catalyzed by  $\text{FeCl}_3$ .

**2. Electrochemistry**

**A. Potentiometry:**

- Determination of redox potential of  $\text{Fe}^{2+}/\text{Fe}^{3+}$  by potentiometric titration of ferrous ammonium sulphate vs potassium dichromate.
- Precipitation titration of  $\text{KCl}$  vs  $\text{AgNO}_3$  –Determination of given concentration of silver nitrate.

**B. pH metry:**

- pH metric titration of strong acid ( $\text{HCl}$ ) vs strong base- Determination of the concentration of given acid.
- pH metric titration of strong acid (acetic acid) with strong base ( $\text{NaOH}$ )- Determination of acid dissociation constant ( $K_a$ ) of weak acid.

**3. Conductometry:**

- Determination of overall order: Saponification of ethyl acetate with  $\text{NaOH}$  by conductance measurement

**Reference books:**

- Senior practical physical chemistry, B.D.Khosla, V.C.Garg, Adarsh Guati.
- Advanced Practical Physical chemistry, J.B.Yadav.
- Practical Physical chemistry, B.Vishvanathan and P.S.Raghavan.
- Practical Physical chemistry, P.S. Sindhu.

